Research Paper Entries

Title – problem statement

Explainable Financial Fraud Detection Using Graph Neural Networks: A Comparative Study

Abstract

Introduction

Literature – whatever has been done on that topic till now

Methodology / proposed architecture – what new am I doing?

1. **Data set**
2. **Methodology**

Result

Conclusion

Image/pics – draw all using draw.io

Caption – below the image and above the table

ROADMAP

Machine Learning

↓

Neural Networks

↓

Deep Learning

↓

Graphs

↓

Message Passing

↓

Graph Convolutions

↓

PyTorch Geometric

↓

GNNExplainer

Skills Required

I've divided them into **Must Know**, **Should Know**, and **Nice to Know**.

Level 1 (Must Know)

Everyone in the team should know these.

1. Python

Need to know

- variables
- loops
- functions
- lists
- dictionaries
- classes (basic)

Difficulty:

⭐☆☆☆☆

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2. NumPy

Need

- arrays
- indexing
- matrix basics

Difficulty

⭐⭐☆☆☆

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3. Pandas

Need

- loading CSV
- filtering
- missing values
- grouping
- statistics

Difficulty

⭐⭐☆☆☆

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4. Machine Learning Basics

Understand

- classification
- training
- testing
- validation
- overfitting
- precision
- recall
- F1-score
- confusion matrix

Difficulty

⭐⭐☆☆☆

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5. Neural Network Basics

Need

- neurons

- hidden layers
- activation functions
- forward propagation
- loss function
- optimizer

Not mathematical derivations.

Difficulty

⭐ ⭐ ⭐ ☆☆

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6. Fraud Detection Concepts

Need to know

- transaction fraud
- account fraud
- money laundering
- fraudulent behavior
- class imbalance

Difficulty

⭐ ⭐ ☆☆☆

Level 2 (Project Skills)

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7. Graph Theory

Need

- nodes

- edges
- neighbors
- directed graph
- weighted graph

Don't study advanced graph algorithms.

Difficulty

⭐ ⭐ ⭐ ☆☆

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8. Graph Neural Networks

Need

- Why graphs?
- Why GNN instead of CNN?
- Message Passing
- Node embeddings
- Graph embeddings

Know these models:

- GCN
- GraphSAGE
- GAT

Don't build them yourself.

Just use PyTorch Geometric implementations.

Difficulty

⭐ ⭐ ⭐ ⭐ ☆

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9. Deep Learning Framework

PyTorch

Need

- tensors
- Dataset
- DataLoader
- model training
- optimizer

Difficulty

⭐⭐⭐☆☆

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10. PyTorch Geometric

Need

- Data object
- Graph creation
- Train GCN
- Train GraphSAGE
- Train GAT

Difficulty

⭐⭐⭐⭐☆

Level 3 (Explainable AI)

Need

Explainability

Understand

Why did model predict fraud?

Methods

- GNNExplainer
- Integrated Gradients (optional)
- Attention visualization (for GAT)

Difficulty

⭐ ⭐ ⭐ ☆☆

Level 4 (Research)

Need

- reading papers
- literature survey
- citation
- comparison tables
- writing methodology
- writing results

Difficulty

⭐ ⭐ ☆☆☆

Libraries You'll Use

Python

↓

NumPy

↓

Pandas

↓

Scikit-learn

↓

PyTorch

↓

PyTorch Geometric

↓

NetworkX

↓

Matplotlib

↓

GNNExplainer

That's all. No TensorFlow is necessary.

Work Distribution (4 Members)

This division keeps everyone involved while minimizing overlap.

Member 1 — Literature & Research Lead

Responsibilities:

- Collect and read research papers
- Write Introduction
- Write Literature Review
- Manage references and citations
- Compare previous work

Skills:

- Research
- Academic writing

Member 2 — Data & Preprocessing Lead

Responsibilities:

- Find and understand datasets
- Clean data
- Create graph structure (nodes and edges)
- Perform exploratory data analysis (EDA)
- Prepare data for training

Skills:

- Pandas
- NumPy
- NetworkX

- Data preprocessing

Member 3 — Model Development Lead

Responsibilities:

- Implement GCN, GraphSAGE, and GAT using PyTorch Geometric
- Train and evaluate models
- Tune basic hyperparameters
- Record performance metrics

Skills:

- PyTorch
- PyTorch Geometric
- Machine learning evaluation

Member 4 — Explainability & Evaluation Lead

Responsibilities:

- Apply GNNExplainer
- Visualize explanations
- Compare model interpretability
- Create graphs, charts, and tables
- Write Results, Discussion, and Conclusion
- Help prepare the final presentation

Skills:

- GNNExplainer
- Matplotlib
- Academic presentation

Suggested Timeline (8 Weeks)

1	Python, NumPy, Pandas, ML basics
2	Neural networks and PyTorch fundamentals
3	Graph theory and NetworkX
4	Learn GCN, GraphSAGE, and GAT concepts
5	Dataset preparation and graph construction
6	Train models and collect evaluation metrics
7	Apply GNNExplainer, compare results, create visualizations
8	Write the research paper, finalize presentation, and rehearse

My Recommendation for Your Team

Since your team mentioned wanting to keep the project manageable, I would **avoid implementing GNN algorithms from scratch**. Instead:

- Use an existing financial fraud dataset (such as IEEE-CIS Fraud Detection or Elliptic Bitcoin Transactions, depending on project scope).
- Build graphs from the transaction data.
- Train three well-known GNN models (GCN, GraphSAGE, and GAT) using **PyTorch Geometric**.
- Compare their fraud detection performance and explainability using **GNNExplainer**.
- Focus your research contribution on the **comparative analysis** of accuracy, efficiency, and interpretability rather than proposing a new algorithm.

This approach is technically solid, feasible for an undergraduate team, and aligns well with your title while leaving enough depth for a strong research paper.